eigenspace decomposition of C must be a decomposition of representations).

(i) Show that V has a subrepresentation $W = V_\lambda$ such that $V/W = nV_\lambda$ for some n (use (h) and the fact that V is the smallest reducible representation which cannot be decomposed).

(j) Deduce from (i) that the eigenspace $V(\lambda)$ of H is $(n+1)$-dimensional. If $v_1, \dots, v_{n+1}$ is its basis, show that $F^j v_i$, $1 \le i \le n+1$, $0 \le j \le \lambda$, are linearly independent and therefore form a basis of V (establish that if $Fx = 0$ and $Hx = \mu x$ for $x \neq 0$, then $Cx = \frac{\mu(\mu-2)}{2}x$ and hence $\mu = -\lambda$).

(k) Define $W_i = \mathrm{span}(v_i, Fv_i, \dots, F^\lambda v_i)$. Show that W_i are subrepresentations of V and derive a contradiction to the fact that V cannot be decomposed.

(l) (Jacobson-Morozov lemma) Let V be a finite dimensional complex vector space and $A : V \to V$ a nilpotent operator. Show that there exists a unique, up to an isomorphism, representation of $\mathfrak{sl}(2)$ on V such that $E = A$. (Use the classification of the representations and the Jordan normal form theorem.)

(m) (Clebsch-Gordan decomposition) Find the decomposition of the representation $V_\lambda \otimes V_\mu$ of $\mathfrak{sl}(2)$ into irreducibles components.

Hint: For a finite dimensional representation V of $\mathfrak{sl}(2)$ it is useful to introduce the character $\chi_V(x) = Tr(e^{xH})$, $x \in \mathbb{C}$. Show that $\chi_{V\oplus W}(x) = \chi_V(x) + \chi_W(x)$ and $\chi_{V\otimes W}(x) = \chi_V(x)\chi_W(x)$. Then compute the character of V_λ and of $V_\lambda \otimes V_\mu$ and derive the decomposition. This decomposition is of fundamental importance in quantum mechanics.

(n) Let $V = \mathbb{C}^M \otimes \mathbb{C}^N$ and $A = J_{0,M} \otimes \mathrm{Id}_N + \mathrm{Id}_M \otimes J_{0,N}$, where $J_{0,n}$ is the Jordan block of size n with eigenvalue zero (i.e., $J_{0,n}e_i = e_{i-1}$, $i = 2, \dots, n$, and $J_{0,n}e_1 = 0$). Find the Jordan normal form of A using (l) and (m).

2.16. Problems on Lie algebras

Problem 2.16.1 (Lie's theorem)**.** The **commutant** $K(\mathfrak{g})$ of a Lie algebra $\mathfrak{g}$ is the linear span of elements $[x, y]$, $x, y \in \mathfrak{g}$. This is an ideal in $\mathfrak{g}$ (i.e., it is a subrepresentation of the adjoint representation). A